

Food Bank Surplus Redistribution Platform

JUSTIFICATION

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Architecture Selection

We chose Microservices Architecture for the Food Bank Surplus Redistribution Platform.

Reason 1 – Independent Services:

The platform has different functions such as food inventory and expiry tracking, order/reservation management, and notifications. Microservices allow these functions to operate as separate services, making the system easier to maintain and modify independently.

Reason 2 – Fast Notifications and Scalability:

The system must send instant notifications to shelters within a 5 km radius when high-quantity surplus food is listed. A separate Notification Service can handle these requests independently and can be scaled when the number of notifications increases.

Security Advantage:

Microservices allow access to different services to be controlled through separate APIs. This helps restrict unauthorized access and provides better separation between system functions.

Performance Benefit:

Individual services can operate and scale independently. For example, the Notification Service can handle a large number of shelter notifications without requiring the entire platform to be scaled.